Practice assignment 9 - Not Graded

This assignment will not be graded and is only for practice.

Multiple Select Question (MSQ)

1 point

Consider a function $f: R^2 \rightarrow R$f : R2 → R defined as:

$$f(x, y) = \begin{cases} \frac{2xy}{\sqrt{2(x^2+y^2)}} & \text{if } (x, y) \neq (0, 0) \\ 1 & \text{if } (x, y) = (0, 0). \end{cases}$$ f(x, y) = { 2(x2+y2)

√

2xy1if (x, y) = (0, 0)if (x, y) = (0, 0).

Choose the set of correct options.

- [ ] $|x| \leq \sqrt{x^2+y^2}$|x| ≤ x2 + y2

√

and $|y| \leq \sqrt{x^2+y^2}$|y| ≤ x2 + y2

√

.

- [ ] If $(x, y) \rightarrow (0, 0)$(x, y) → (0, 0), then $f(x, y) \rightarrow 0$f(x, y) → 0.
- [ ] $\lim_{(x,y)\rightarrow(0,0)} f(x, y)$(x,y)→(0,0)lim f(x, y) exists.
- [ ] $\lim_{(x,y)\rightarrow(0,0)} f(x, y) = 0$(x,y)→(0,0)lim f(x, y) = 0.
- [ ] $f(x, y)$f(x, y) is continuous at the origin (0,0).

No, the answer is incorrect.

Score: 0

Accepted Answers:

$|x| \leq \sqrt{x^2+y^2}$|x| ≤ x2 + y2

√

and $|y| \leq \sqrt{x^2+y^2}$|y| ≤ x2 + y2

√

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If $(x, y) \rightarrow (0, 0)$(x, y) → (0, 0), then $f(x, y) \rightarrow 0$f(x, y) → 0.

$\lim_{(x,y)\rightarrow(0,0)} f(x, y)$(x,y)→(0,0)lim f(x, y) exists.

$\lim_{(x,y)\rightarrow(0,0)} f(x, y) = 0$(x,y)→(0,0)lim f(x, y) = 0.

1 point

Consider the following functions

$$f(x, y) = (\sqrt{1-x}, \sqrt{y})$$ f(x, y) = (1 − x

√

, y
√
)

$$g(x, y) = \sin((x^2 - 1)^2 + y^4)$$ g(x, y) = sin ((x2 − 1)2 + y4)

$$h(x, y) = \frac{x}{x^2 + y^2}$$ h(x, y) = x2+y2x

$$s(x, y) = x^2 e^{y^2}$$ s(x, y) = x2ey2

Which of the following is true?

☐

Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}$ f = {(x, y) | x ≤ 1, y ≥ 0}

☐

Domain of $h = R^2$ h = R2

☐

$g \circ f(x, y) = \sin(x^2 - 1 + y^2)$ g ∘ f(x, y) = sin (x2 − 1 + y2)

☐

$((g \circ f) \times h)(x, y) = \frac{\sin(x^2 + y^2)}{x^2 + y^2}$ ((g ∘ f) × h)(x, y) = x2+y2sin (x2+y2)

☐

$\lim_{(x, y) \to (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$ (x,y)→(0,0)lim((g ∘ f) × h + s)(x, y) = 0

No, the answer is incorrect.
Score: 0
Accepted Answers:
Domain of $f = \{(x, y) \mid x \leq 1, y \geq 0\}$ f = {(x, y) | x ≤ 1, y ≥ 0}
$\lim_{(x, y) \to (0, 0)} ((g \circ f) \times h + s)(x, y) = 0$ (x,y)→(0,0)lim((g ∘ f) × h + s)(x, y) = 0

1 point

Match the functions of two variables in Column A with their graphs given in Column B.

☐

i → → 4, ii → → 3

☐

i → → 3, ii → → 4

☐

iii → → 1, iv → → 2

☐

iii → → 2, iv → → 1

No, the answer is incorrect.
Score: 0
Accepted Answers:
i → → 3, ii → → 4
iii → → 1, iv → → 2

1 point

Consider two functions $f: R^2 \to R^2$ and $g: R^2 \to R$ f : R2 → R2 and g : R2 → R defined as:

$$f(x, y) = (x^2, y^2)$$ f(x, y) = (x2, y2)

and

$g(x, y) = \{\frac{x^2 - y^2}{x^2 + y^2} \text{if } (x, y) \neq (0, 0)$ g(x, y) = {x2+y2x2−y20if (x, y) = (0, 0)if (x, y) = (0, 0)
$0 \quad \text{if } (x, y) = (0, 0)$

Which of the following option(s) is (are) true?

- [ ] $g \circ f(x, y) = \{\frac{x^2 - y^2}{x^2 + y^2} \text{if } (x, y) \neq (0, 0)$ g ∘ f(x, y) = {x2+y2x2−y20if (x, y) = (0, 0)if (x, y) = (0, 0)
$0 \quad \text{if } (x, y) = (0, 0)$
- [ ] $g \circ f(x, y) = \{\frac{x^4 - y^4}{x^4 + y^4} \text{if } (x, y) \neq (0, 0)$ g ∘ f(x, y) = {x4+y4x4−y40if (x, y) = (0, 0)if (x, y) = (0, 0)
$0 \quad \text{if } (x, y) = (0, 0)$
- [ ] $\lim_{(x, y) \to (0, 0)} g(x, y)$(x,y)→(0,0)lim g(x, y) exists.
- [ ] $\lim_{(x, y) \to (0, 0)} g \circ f(x, y)$(x,y)→(0,0)lim g ∘ f(x, y) exists.
- [ ] $\lim_{x \to 0} \lim_{y \to 0} g \circ f(x, y) = \lim_{y \to 0} \lim_{x \to 0} g \circ f(x, y)$x→0lim y→0lim g ∘ f(x, y) = y→0lim x→0lim g ∘ f(x, y)
- [ ] $\lim_{x \to 0} \lim_{y \to 0} g(x, y) = 1$x→0lim y→0lim g(x, y) = 1
- [ ] $\lim_{y \to 0} \lim_{x \to 0} g \circ f(x, y) = -1$y→0lim x→0lim g ∘ f(x, y) = −1
- [ ] $\lim_{(x, y) \to (1, 1)} g(x, y)$(x,y)→(1,1)lim g(x, y) does not exist.

No, the answer is incorrect.
Score: 0
Accepted Answers:
$g \circ f(x, y) = \{\frac{x^4 - y^4}{x^4 + y^4} \text{if } (x, y) \neq (0, 0)$ g ∘ f(x, y) = {x4+y4x4−y40if (x, y) = (0, 0)if (x, y) = (0, 0)
$0 \quad \text{if } (x, y) = (0, 0)$
$\lim_{x \to 0} \lim_{y \to 0} g(x, y) = 1$x→0lim y→0lim g(x, y) = 1
$\lim_{y \to 0} \lim_{x \to 0} g \circ f(x, y) = -1$y→0lim x→0lim g ∘ f(x, y) = −1

Numerical Answer Type (NAT)
Consider a function $f(x, y) = xy$f(x, y) = xy. If the directional derivative of ff at point (a, b)(a, b), in direction of (1,-1) and (1,1) are 1 and 5 respectively. Then find the value $\frac{2a + b}{\sqrt{2}}$ 2
$\sqrt{}$
2a+b.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 7
1 point
Consider the following statements:

Statement 1: Let $f: R^2 \rightarrow R$f : R2 → R be a function such that $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f$x→alim y→blim f exists, where $a, b \in R$a, b ∈ R. Then $\lim_{(x, y) \rightarrow (a, b)} f$(x,y)→(a,b)lim f exists.

Statement 2: Let $f: R^2 \rightarrow R$f : R2 → R be a function such that $\lim_{y \rightarrow b} \lim_{x \rightarrow a} f = f(a, b)$y→blim x→alim f = f(a, b), where $a, b \in R$a, b ∈ R. Then ff is continuous.

Statement 3: Let $f: R^2 \rightarrow R$f : R2 → R be a function such that $\lim_{x \rightarrow a} \lim_{y \rightarrow b} f = \lim_{y \rightarrow b} \lim_{x \rightarrow a} f = f(a, b)$ x→alim y→blim f = y→blim x→alim f = f(a, b), where $a, b \in R$a, b ∈ R. Then ff is continuous.

Statement 4: Let $f: R^2 \rightarrow R$f : R2 → R be a function such that $f_x(a, b)$fx(a, b) exists. Then ff is continuous.

Statement 5: Let $f: R^2 \rightarrow R$f : R2 → R be a function such that $f_y(a, b)$fy(a, b) exists. Then ff is continuous.

How many statements are correct?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: String) 0
1 point

Consider the following two functions $f_1, f_2: R^2 \rightarrow R$f1, f2 : R2 → R defined as:

$$f_1(x, y) = xy + 2x \text{f1(x, y) = xy + 2x}$$
$$f_2(x, y) = x^2 + 2y + 3x^2 y \text{.f2(x, y) = x2 + 2y + 3x2y.}$$

Define a matrix AA as follows:

$$A = [\frac{\partial f_1}{\partial x}(3, 0) \frac{\partial f_2}{\partial x}(1, 1) \frac{\partial f_1}{\partial y}(3, 2) \frac{\partial f_2}{\partial y}(\sqrt{\frac{7}{3}}, 2)] \text{.A = [∂x∂f1(3, 0)∂y∂f1(3, 2)∂x∂f2(1, 1)∂y∂f2 (} \sqrt{37}, 2)] \text{.}$$

What will be the value of $det(A)$det(A)?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -6
1 point

Comprehension Type Question

An earthworm is crawling about on a sunny day. We draw coordinates axes on the ground where it is crawling and find that the body temperature of the earthworm at a point (x, y)(x, y) is given by the function:

$$T(x, y) = 2x^2 + 3xy + y^2 \text{T(x, y) = 2x2 + 3xy + y2}$$

Use this information to answer questions 8,9, and 10.

1 point

Which of the following option(s) is (are) true?

☐ $\lim_{(x,y)\to(1,1)} T(x,y) = \lim_{x\to 1}\lim_{y\to 1} T(x,y)$(x,y)→(1,1)lim T (x, y) = x→1lim y→1lim T (x, y)

☐ $T(x,y)$T (x, y) is continuous at (1,1)

☐ $T(x,y)$T (x, y) is a linear function.

☐ $T(cx,cy) = c\,T(x,y)$T (cx, cy) = c T (x, y).

☐ If the earthworm approaches the point $(1,2)$(1, 2) from any direction, then the body temperature of the earthworm approaches 12 units.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim_{(x,y)\to(1,1)} T(x,y) = \lim_{x\to 1}\lim_{y\to 1} T(x,y)$(x,y)→(1,1)lim T (x, y) = x→1lim y→1lim T (x, y)

$T(x,y)$T (x, y) is continuous at (1,1)

If the earthworm approaches the point $(1,2)$(1, 2) from any direction, then the body temperature of the earthworm approaches 12 units.

If the gradient of $T(x,y)$T (x, y) is (25, 18) at a point, then find the body temperature of the earthworm at that point.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 77

1 point

1 point

Which of the following option(s) is (are) true?

☐ If the earthworm moves parallel to the $X-$ X−axis, then rate of change of its body temperature at the point (a,b)(a, b) is given by $4a+3b$4a + 3b.

☐ If the earthworm moves parallel to the $Y-$ Y −axis, then rate of change of its body temperature at the point (a,b)(a, b) is given by $2a+3b$2a + 3b.

☐ If the earthworm moves parallel to the straight line joining the origin and the point $(2,3)$(2, 3), then rate of change of its body temperature at the point $(1,1)$(1, 1) is given by $\frac{29}{\sqrt{13}}$ 13

√

29.

☐ If the earthworm moves parallel to the straight line joining the origin and the point $(0,1)$(0, 1), then rate of change of its body temperature at the point $(1,2)$(1, 2) is given by 1010.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If the earthworm moves parallel to the $X-$ X−axis, then rate of change of its body temperature at the

point (a, b)(a, b) is given by $4a$ + $3b$4a + 3b.
If the earthworm moves parallel to the straight line joining the origin and the point $(2, 3)$(2, 3), then rate of change of its body temperature at the point $(1, 1)$(1, 1) is given by $\frac{29}{\sqrt{13}}$ 13

$\sqrt{}$
29.

Check Answers